An Kinetic Theory Based Investigation Into The Dispersion Relations And Plasma Oscillations in Bound Maxwellian 2D Plasmas Subject to 3D Electrostatic Fields

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Methods

Assume a 2D plasma bound along the xy plane and let z be the reference axis such that $\vec{v} = v_x\hat{x} + v_y\hat{y}$. Due to the nature of the plasma sheet , it is assumed to follow the Maxwell distribution and the ions are assumed fixed. This derivation was done assuming electrostatic fields and no magnetic fields. The distribution function of this constrained planar plasma has the form of equation [1]. Note that α represents the species

$$f(\vec{v}_\perp) = n(\vec{x},t)\,\hat{f}(\vec{v}_\perp)\delta(z)\delta(v_z) \text{ [1]}$$

Where $f(\vec{v}_\perp)$ is the 2D Maxwell distribution given by equation [2].

$$\hat{f}(\vec{v}_\perp) = \frac{m_\alpha}{2\pi k_B T_\alpha} exp(\frac{-m_\alpha v_\perp^{\ 2}}{2k_B T_\alpha}) \text{ [2]}$$

The initial distribution of the plasma may be modeled as unperturbed by oscillations,

$$f_0(\vec{v}_\perp) = n_o(\vec{x},t)\,\hat{f}_o(\vec{v}_\perp)\delta(z)\delta(v_z)$$

The vlasov equation for the 2D distribution consequently yields [3]

$$\frac{\partial f(\vec{v}_\perp)}{\partial t}\delta(z)\delta(v_z) + \vec{v}\cdot\frac{\partial f(\vec{v}_\perp)}{\partial \vec{x}}\delta(z)\delta(v_z) + \frac{q_\alpha}{m_\alpha}\vec{E}\cdot\frac{\partial f(\vec{v}_\perp)}{\partial \vec{v}}\delta(z)\delta(v_z) = 0 \text{ [3]}$$

Linearization of [3] yielded [4]

$$\frac{\partial f_1(\vec{v}_\perp)}{\partial t} + \vec{v}\cdot\frac{\partial f_1(\vec{v}_\perp)}{\partial \vec{x}} + \frac{q_\alpha}{m_\alpha}\vec{E}_1\cdot\frac{\partial f_o(\vec{v}_\perp)}{\partial \vec{v}} = 0 \text{ [4]}$$

Application of a fourier transform allows this equation to be expressed as [5]

$$f_1(\vec{k}_\perp,\vec{v}_\perp,\omega) = \frac{-iq_\alpha}{m_\alpha\left[\omega-(\vec{v}_\perp\cdot\vec{k}_\perp)\right]}\vec{E}_1\cdot\frac{\partial f_o}{\partial \vec{v}} \text{ [5]}$$

Since only electrons are mobile, the above equations apply only to the electrons and not the ions, as such the alpha notation may be dropped and we may represent this in terms of only electron quantities.

$$f_1(\vec{k}_\perp,\vec{v}_\perp,\omega) = \frac{ie}{m_e\left[\omega-(\vec{v}_\perp\cdot\vec{k}_\perp)\right]}\vec{E}_1\cdot\frac{\partial f_o}{\partial \vec{v}} \text{ [6]}$$

Poisson's equation yields

$$-\nabla^2\phi = \frac{en_{e1}}{\epsilon_0}\delta(z) \Rightarrow \text{fourier transform} \Rightarrow -k_\perp^{\ 2}\phi + \frac{\partial^2\phi}{\partial z^2} = \frac{e}{\epsilon_0}\delta(z)\int\int f_1(\vec{k}_\perp,\vec{v}_\perp,\omega)d^2v$$

$$-k_\perp^{\ 2}\phi + \frac{\partial^2\phi}{\partial z^2} = \frac{e}{\epsilon_0}\delta(z)\int\int\frac{ie}{m_e\left[\omega-(\vec{v}_\perp\cdot\vec{k}_\perp)\right]}\vec{E}_1\cdot\frac{\partial f_o}{\partial \vec{v}}d^2v \Rightarrow k_\perp^{\ 2}\phi - \frac{\partial^2\phi}{\partial z^2} = -\frac{e}{\epsilon_0}\delta(z)\int\int\frac{ie}{m_e\left[\omega-(\vec{v}_\perp\cdot\vec{k}_\perp)\right]}\vec{E}_1\cdot\frac{\partial f_o}{\partial \vec{v}}d^2v$$

$$k_\perp^{\ 2}\phi - \frac{\partial^2\phi}{\partial z^2} = \frac{e^2\phi}{k_B T_e\epsilon_0}\delta(z)\int\int\frac{\vec{k}_\perp\cdot\vec{v}_\perp}{\omega-(\vec{k}_\perp\cdot\vec{v}_\perp)}f_0 d^2v \text{ [7]}$$

All dispersion relations will follow from equation [7].

High Frequency Oscillations

In the case that omega is large, one may assume that $\vec{k}_\perp \cdot \vec{v}_\perp << \omega$ and thus expand the integral as

$\int\int\left[\frac{\vec{k}_\perp\cdot\vec{v}_\perp}{\omega}+\left(\frac{\vec{k}_\perp\cdot\vec{v}_\perp}{\omega}\right)^2+\left(\frac{\vec{k}_\perp\cdot\vec{v}_\perp}{\omega}\right)^3+\left(\frac{\vec{k}_\perp\cdot\vec{v}_\perp}{\omega}\right)^4+..\right]f_0 d^2v \Rightarrow$ Since the distribution is exponential, all odd terms drop.

Keeping only the first two terms we obtained [8]

$$\int\int\left[\left(\frac{\vec{k}_\perp\cdot\vec{v}_\perp}{\omega}\right)^2+\left(\frac{\vec{k}_\perp\cdot\vec{v}_\perp}{\omega}\right)^4\right]f_0 d^2v \text{ [8]}$$

The integration for these is most easily done by expressing $\vec{k}_\perp \cdot \vec{v}_\perp$ as $k_\perp v_\perp cos(\theta)$ and integrating in spherical coordinates

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\left[\left(\frac{\vec{k}_\perp\cdot\vec{v}_\perp}{\omega}\right)^2+\left(\frac{\vec{k}_\perp\cdot\vec{v}_\perp}{\omega}\right)^4\right]f_0 d^2v \Rightarrow \int_0^{2\pi}\int_0^{\infty}\left[\left(\frac{k_\perp v_\perp cos(\theta)}{\omega}\right)^2+\left(\frac{k_\perp v_\perp cos(\theta)}{\omega}\right)^4\right]f_0 d^2v \text{ [9]}$$

Combining [9], [7], and [2] yields a differential equation [9]

$$k_\perp^2\phi - \frac{\partial^2\phi}{\partial z^2} = \frac{n_o e^2}{m_e\epsilon_0}\frac{k_\perp^2}{\omega^2}\phi\delta(z) + \frac{3n_o e^2}{m_e^2\epsilon_0}k_B T_e\frac{k_\perp^4}{\omega^4}\phi\delta(z) \text{ [10]}$$

The solution to this differential equation may be obtained with a fourier transform the details of which are included in the appendix. The quantity $\omega_{p,dep}^2 = \frac{n_o e^2}{m_e\epsilon_0}$ will be referred to henceforth as the "depressed" plasma frequency $\omega_{p,dep}$ due to the units not being 1/s. The result of this fourier transform is then

$$\phi(z) = \frac{\omega_{p,dep}^2}{2\omega^2}k_\perp\phi_0 exp(-k_\perp|z|) + \frac{3\omega_{p,dep}^2}{2m_e\omega^2}k_\perp^3 k_B T_e\phi_0 exp(-k_\perp|z|) \text{ [10]}$$

Which at z = 0 yields the dispersion relation for this plasma

$$\omega^2 = \frac{\omega_{p,dep}^2}{2}k_\perp + \frac{3\omega_{p,dep}^2}{2m_e\omega^2}k_\perp^3 k_B T_e \text{ [11]}$$

1.2)Coupling

In the case that two sheets of these plasmas are placed a distance apart from each other, the coupling affects the dispersion relation. Let a plasma sheet be placed in the xy plane and another sheet be a distance "a" above this sheet as in fig 1. The distribution now exists for two sheets, and as such is represented by two delta functions

$$f_0(\vec{v}_\perp) = n_o(\vec{x},t)\,\hat{f}_o(\vec{v}_\perp)(\delta(z) + \delta(z-a))\delta(v_z) \text{ [12]}$$

The coupled analog of equation [7] (the governing differential equation) can be easily read off as equation [13]. For ease of computability, the derivation will be demonstrated with a cold plasma approximation.

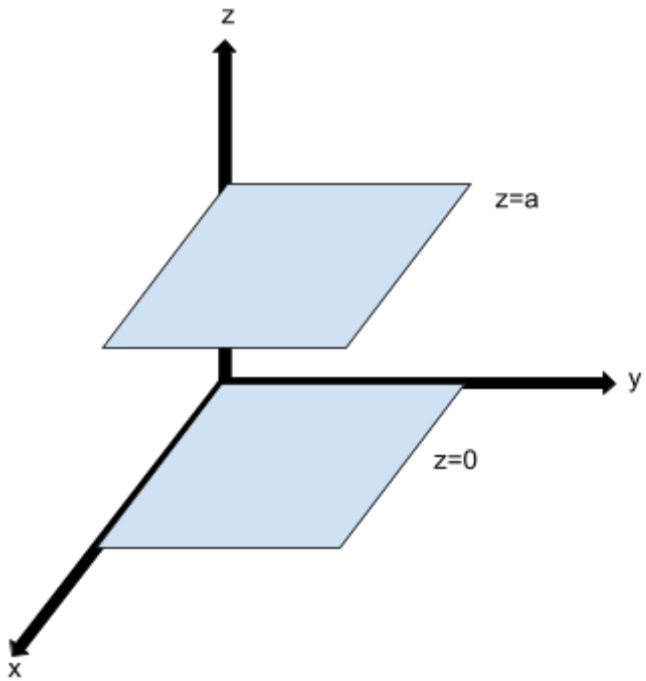

Fig 1. Sketch of coupled plasma distribution

$$k_\perp^2\phi - \frac{\partial^2\phi}{\partial z^2} = \frac{e^2\phi}{k_B T_e \epsilon_0}(\delta(z) + \delta(z-a))\int\int \frac{\vec{k}_\perp \cdot \vec{v}_\perp}{\omega - (\vec{k}_\perp \cdot \vec{v}_\perp)} f_0 d^2v \text{ [13]}$$

Using the previously developed integral, equation [13] becomes

$$k_\perp^2\phi - \frac{\partial^2\phi}{\partial z^2} = \frac{e^2\phi}{k_B T_e \epsilon_0}(\delta(z) + \delta(z-a))\left[\frac{n_0 k_b T_e}{m_e \omega^2}k_\perp^2 + \frac{3n_0(k_B T_e)^2}{m_e^2\omega^4}k_\perp^4\right] \Rightarrow cold\ plasma \Rightarrow$$

$$k_\perp^2\phi - \frac{\partial^2\phi}{\partial z^2} = \frac{\omega_{p,dep}^2}{\omega^2}k_\perp^2\phi(\delta(z) + \delta(z-a)) \text{ [15]}$$

Equation [15] may be solved with a fourier transform similar to equation [10], the details will again be included in the appendix. The solution to [15] is [16]

$$\phi(z) = \frac{\omega_{p,dep}^2}{2\omega^2}k_\perp\phi_0 exp(-k_\perp|z|) + \frac{\omega_{p,dep}^2}{2\omega^2}k_\perp\phi_a exp(-k_\perp|z-a|) \text{ [16]}$$

This decomposes into the following equation system

$$\phi_0 = \frac{\omega_{p,dep}^2}{2\omega^2}k_\perp\phi_0 + \frac{\omega_{p,dep}^2}{2\omega^2}k_\perp\phi_a exp(-k_\perp a)$$

$$\phi_a = \frac{\omega_{p,dep}^2}{2\omega^2}k_\perp\phi_0 exp(-k_\perp a) + \frac{\omega_{p,dep}^2}{2\omega^2}k_\perp\phi_a$$

The system may be represented as a matrix and its eigenvalue taken, when the eigenvalue $\lambda = 1$, the nontrivial solution falls out and allows for the determination of the dispersion relation. The system is

$$\begin{pmatrix}\phi_0\\ \phi_a\end{pmatrix} = \frac{\omega_{dep}^2}{2\omega^2}K_\perp\begin{pmatrix}1 & exp(-K_\perp a)\\ exp(-K_\perp a) & 1\end{pmatrix}\begin{pmatrix}\phi_0\\ \phi_a\end{pmatrix}$$

with eigenvalues

$\lambda_1 = \frac{\omega_{p,dep}^2}{2\omega^2} k_\perp \left(1 + exp(-k_\perp a)\right)$ and $\lambda_2 = \frac{\omega_{p,dep}^2}{2\omega^2} k_\perp \left(1 - exp(-k_\perp a)\right)$ with the following eigenfrequencies falling out: giving us the dispersion relation for the coupled plasma sheets.

$$\omega_1^2 = \frac{\omega_{p,dep}^2}{2} k_\perp \left(1 + exp(-k_\perp a)\right) \quad [17]$$

$$\omega_2^2 = \frac{\omega_{p,dep}^2}{2} k_\perp \left(1 - exp(-k_\perp a)\right) \quad [18]$$

Doing this with a warm plasma follows identical machinery, the warm coupled plasma dispersion relation is given as

$$1 + \frac{3k_B T_e}{m_e \omega_2^2} k_\perp^2 = \frac{2\omega_2^2}{\omega_{p,dep}^2 k_\perp} + exp(-k_\perp a)\left[1 + \frac{3k_B T_e}{m_e \omega_2^2} k_\perp^2\right] \quad [19]$$

$$1 + \frac{3k_B T_e}{m_e \omega_2^2} k_\perp^2 = \frac{2\omega_2^2}{\omega_{p,dep}^2 k_\perp} - exp(-k_\perp a)\left[1 + \frac{3k_B T_e}{m_e \omega_2^2} k_\perp^2\right] \quad [20]$$

For the high and low frequency coupled thermal modes.

Results

Dispersion relations were plotted for both the warm coupled and uncoupled plasma sheets as seen in figures 2 and 3.

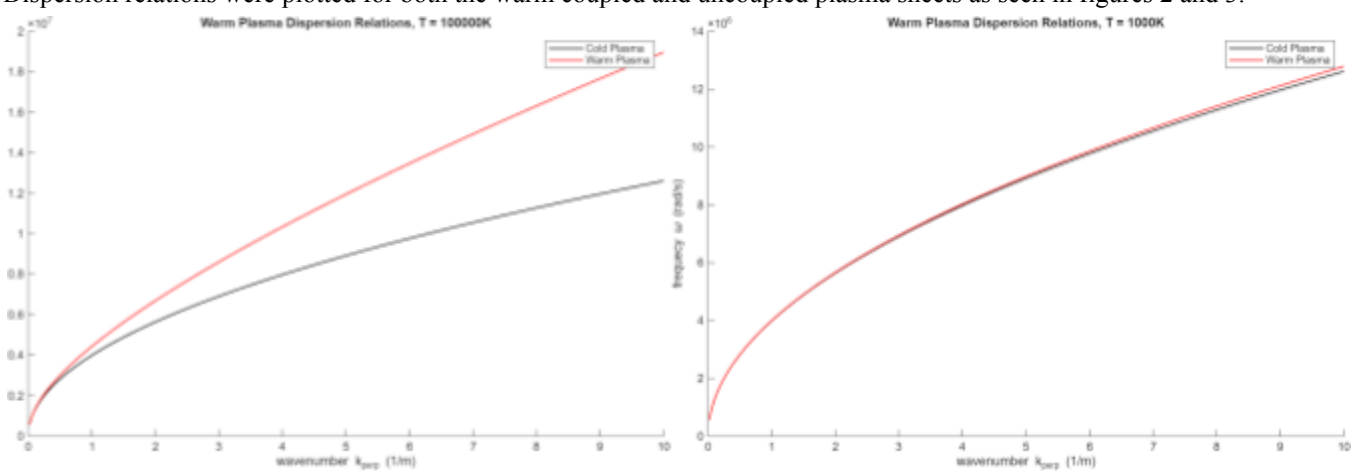

Fig 2. The uncoupled 2D plasma dispersion relation is shown, with the red line representing a warm plasma at T=100000K and T = 1000K respectively, and the black line representing the dispersion relation at T = 0K. The warm dispersion relation converges with the cold dispersion relation as the plasma temperature approaches 0.

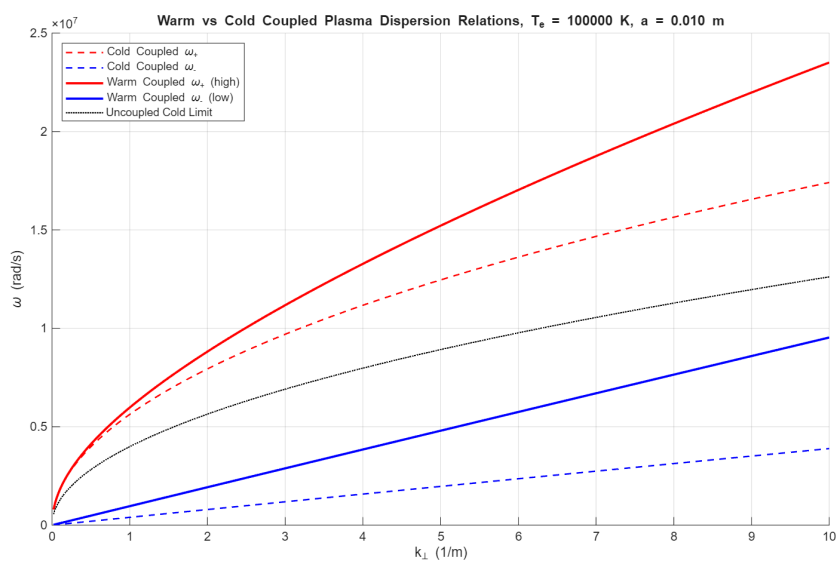

Fig 3. The coupled 2D plasma dispersion relation is shown. The red and blue solid lines correspond to the high frequency and low frequency

warm dispersion relations respectively. The red and blue dotted lines represent the high and low frequency dispersion relations when the plasma is cold. The dotted black line corresponds to the uncoupled plasma limit.

Group velocities for both warm coupled and uncoupled modes were calculated from differentiating the dispersion relations following the definition $v_g = \frac{\partial \omega}{\partial k}$. Cold and warm group velocities were plotted against separation distance 'a' as seen in figure 4.

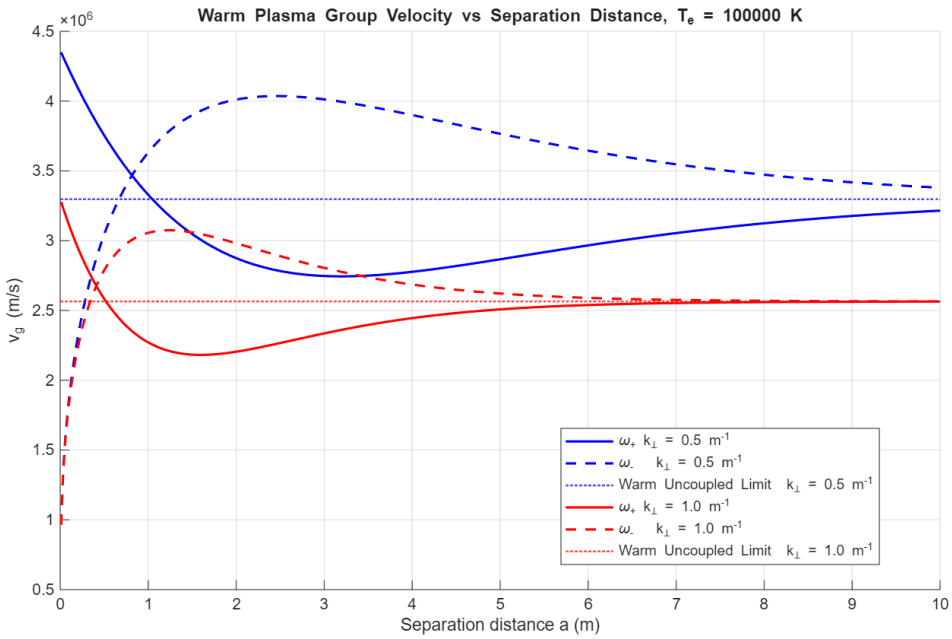

Fig 4. The group velocities are plotted as a function of separation distance parameterized by wavenumber. The low and high frequency modes group velocities converge towards the uncoupled limit as separation distance increases.

Group velocities were also plotted across temperature and wavenumber as seen in fig 5 and 6 for uncoupled sheets.

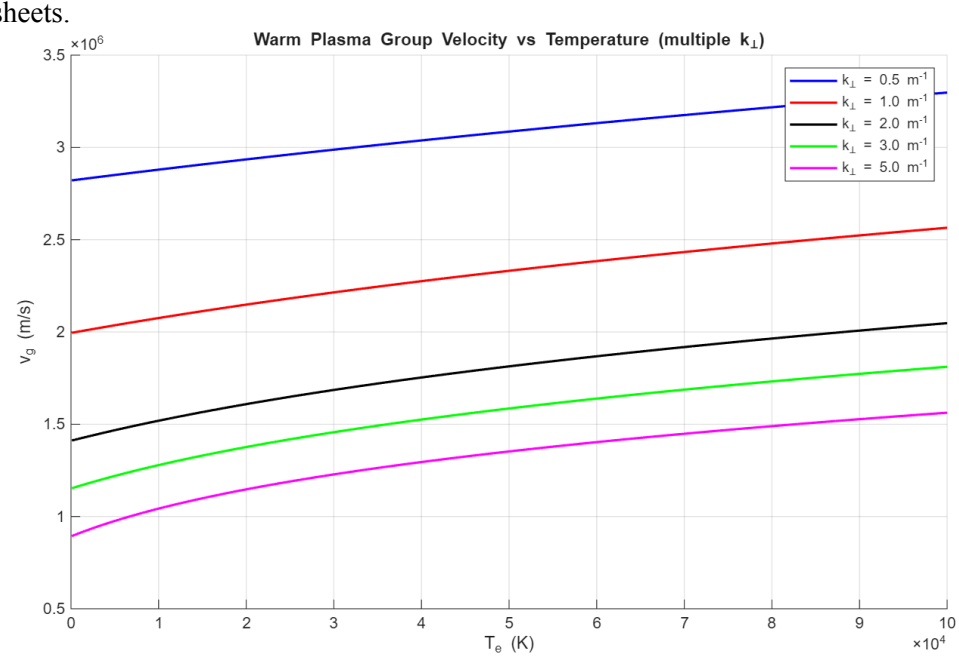

Fig 5. Group velocity vs temperature parametrized by wavenumber. The group velocities are seen to be proportional to the temperature.

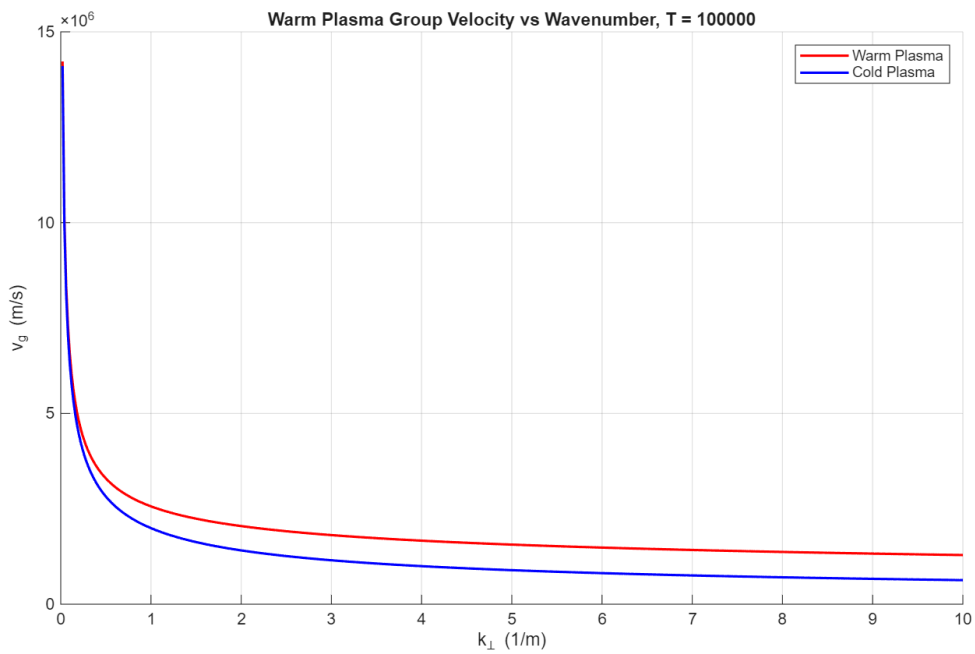

Fig 6. Group velocity vs wavenumber parametrized for warm and cold plasmas. As shown, warm plasmas have higher group velocities relative to cold plasmas across wavenumbers investigated. The effect of temperature on group velocity appears to be more pronounced at higher wavenumbers.

Discussion

The differences between the uncoupled 2D dispersion relation and the normal plasma dispersion relation are vast and immediately clear upon inspection of the dispersion relation proposed in [11]. For a regular plasma, the dispersion relation derived from kinetic theory reads as [21] [4].

$$\omega^2 = \omega_p^2 + \frac{3k^2 k_B T_e}{m_e} \text{ [21]}$$

As seen, the plasma dispersion relation for a normal plasma indicates that ω^2 goes as k^2 for the thermal mode only, and its cold oscillation mode is simply the plasma frequency. This is in stark contrast to the 2D uncoupled plasma dispersion relation, where ω^2 goes as k for the cold mode and as k^3 for the first thermal mode, and both modes depend on the depressed plasma frequency: in contrast to the normal plasma, where only the thermal mode carries a k dependence. This seems to suggest that for a regular plasma, without the presence of thermal motion, oscillations cannot disperse and there is no group velocity, whereas in 2D plasmas it appears that oscillations disperse even in the absence of temperature: meaning that even cold oscillation modes are capable of propagating and transferring wave energy throughout the bulk plasma from an initial perturbation. This results from the geometric nature of the electrostatic interaction: although the charges are confined to the 2D plane, the electric field they generate propagate in 3D. The Fourier transform of the Coulomb potential for a 2D charge distribution introduces an explicit k dependence into the restoring force even in the cold limit, as the out of plane potential directly couples the wavenumber to the electrostatic restoring force. This causes the dispersion relation to retain k dependence even at zero temperature. In other words, there will always be some component of the

perturbed electric field produced by the plasma oscillations pointing in the oscillation plane, allowing for the propagation of waves through energetic interactions with this electric field component, thus causing oscillations to disperse: though the exact nature of this 0K dispersion phenomena will require further experimentation to fully understand and categorize.

The dispersion relation appears more sensitive to thermal effects at higher wavenumbers, as seen in figure 2. At low wavenumbers, the warm and cold dispersion relations overlap closely and differ little, whereas at higher wavenumbers the two diverge drastically. Unsurprisingly, the greater the temperature, the greater this divergence: this is seen in the closer agreement between warm and cold curves for the 1000 K plasma compared to the more pronounced separation in the 10000 K plasma in figure 2. The group velocity is also temperature dependent, as seen in figure 5. The group velocity across a range of wavenumbers show variation with temperature, with higher temperatures yielding higher group velocities throughout. This suggests that oscillations will disperse and propagate faster throughout a warmer plasma than a colder plasma. Physically, this must mean the thermal motion of the charged particles aids in the dispersion of wave energy throughout the bulk 2D plasma, as is consistent with previous work [2].

Coupling effects split the dispersion relation into two distinct modes, as seen in equations [19] and [20]: one high frequency and one low frequency. This arises from the oscillations in the two sheets either constructively or destructively interfering: the high frequency mode corresponds to the two sheets oscillating in phase, reinforcing one another and raising the frequency, while the low frequency mode corresponds to out-of-phase oscillation, suppressing it. As seen in figure 3, the high frequency branch sits above the uncoupled dispersion relation, indicating that at identical wavenumbers the coupled mode oscillates faster, while the low frequency branch sits below, indicating a reduction in frequency at the same wavenumber. The coupling relation between the sheets also varies as an exponential of the separation distances, matching previous literature [5].

Interestingly, at certain separation distances the group velocities of the low and high frequency modes intersect and eventually exchange positions, with the group velocity of the low frequency mode surpassing that of the high frequency mode, as seen in figure 4. This is because the derivatives of the dispersion relation are themselves functions of k as seen in figure 3, and can thus equal in magnitude and crossover. As the sheet separation approaches infinity, however, the group velocities of both modes expectedly converge to the uncoupled limit, as the intersheet coupling vanishes. This crossover effect has been previously observed in literature and can be attributed to a myriad of instabilities [3], specifically the two stream instability which is incredibly prominent in the case of coupled plasma sheets. The lack of investigation into such instabilities and their propagation remains a limitation in the present study.

Lower wavenumber oscillations appear to have greater group velocities as seen in fig 6 and 4 and are also able to sustain coupling effects at greater separation distances, as evidenced by the convergence to the uncoupled limit occurring at larger separation distances for $k = 0.5\ m^{-1}$ compared to $k = 1\ m^{-1}$, seen again in figure 4. Physically, this means that low wavenumber oscillations spread out much faster in 2d plasmas and are able to influence the behavior of sheets at longer distances away from the perturbed sheet. This is not surprising, as long-wavelength electrostatic perturbations are well known to be able to exchange energy at large distances and carry high group velocities [1,6]. The ability for long wavelength oscillations to perturb plasma sheets at greater distances may implicate them in the induction of instabilities in

neighboring plasma sheets and thus, it is important to note coupling effects for future investigations into long wavelength plasma oscillations.

Conclusion

The dispersion relations for uncoupled and coupled 2D plasma sheets were derived from the Vlasov-Poisson system under both cold and warm plasma approximations, with group velocities obtained analytically via implicit differentiation. The resulting relations reveal that 2D confinement introduces a wavenumber dependence absent in bulk plasmas, thermal effects enhance group velocities most strongly at high wavenumbers, and inter-sheet coupling splits the dispersion into two modes whose group velocities converge to the uncoupled limit at large separation distances.

Acknowledgments

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Appendix

A1) Integrals

In the case that $\vec{k}_\perp \cdot \vec{v}_\perp << \omega$

$$\int\int \frac{\vec{k}_\perp \cdot \vec{v}_\perp}{\omega - (\vec{k}_\perp \cdot \vec{v}_\perp)} f_0 d^2v \approx \frac{n_0 k_b T_e}{m_e \omega^2} k_\perp^{\ 2} + \frac{3n_0 (k_B T_e)^2}{m_e^{\ 2} \omega^4} k_\perp^{\ 4} \qquad \int_{-\infty}^{\infty} \frac{k_\perp^{\ 2}}{k_\perp^{\ 2} + \lambda^2} exp(i\lambda z) d\lambda = \pi k_\perp exp(-k_\perp |z|)$$

A2) Fourier Transforms

To solve the differential equation proposed in [10], a fourier transform was applied to it and the inverse was then taken to recover the function in real space.

$$k_\perp^{\ 2}\phi - \frac{\partial^2 \phi}{\partial z^2} = \frac{n_o e^2}{m_e \epsilon_0} \frac{k_\perp^{\ 2}}{\omega^2} \phi \delta(z) + \frac{3n_o e^2}{m_e^{\ 2} \epsilon_0} k_B T_e \frac{k_\perp^{\ 4}}{\omega^4} \phi \delta(z) \Rightarrow$$

$$k_\perp^{\ 2}\phi + \lambda^2 \phi = \frac{n_o e^2}{m_e \epsilon_0} \frac{k_\perp^{\ 2}}{\omega^2} \phi_o + \frac{3n_o e^2}{m_e^{\ 2} \epsilon_0} k_B T_e \frac{k_\perp^{\ 4}}{\omega^4} \phi_o \Rightarrow$$

$$(k_\perp^{\ 2} + \lambda^2)\phi = \frac{n_o e^2}{m_e \epsilon_0} \frac{k_\perp^{\ 2}}{\omega^2} \phi_o + \frac{3n_o e^2}{m_e^{\ 2} \epsilon_0} k_B T_e \frac{k_\perp^{\ 4}}{\omega^4} \phi_o \Rightarrow$$

$\phi = \frac{n_o e^2}{m_e \epsilon_0} \frac{k_\perp^{\ 2}}{\omega^2 (k_\perp^{\ 2} + \lambda^2)} \phi_o + \frac{3n_o e^2}{m_e^{\ 2} \epsilon_0} k_B T_e \frac{k_\perp^{\ 4}}{\omega^4 (k_\perp^{\ 2} + \lambda^2)} \phi_o \Rightarrow$ To recover the function phi in real space

$\phi = \frac{n_o e^2}{m_e \epsilon_0 \omega^2} \phi_o \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{k_\perp^{\ 2}}{(k_\perp^{\ 2} + \lambda^2)} exp(i\lambda z) d\lambda + \frac{3n_o e^2}{m_e^{\ 2} \epsilon_0 \omega^4} k_B T_e \phi_0 \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{k_\perp^{\ 4}}{(k_\perp^{\ 2} + \lambda^2)} exp(i\lambda z) d\lambda \Rightarrow$ from the integral chart, this leads to $\phi(z) = \frac{\omega_{p,dep}^{\ 2}}{2\omega^2} k_\perp \phi_0 exp(-k_\perp |z|) + \frac{3\omega_{p,dep}^{\ 2}}{2m_e \omega^2} k_\perp^{\ 3} k_B T_e \phi_0 exp(-k_\perp |z|)$ which is equation [10].

Equation [15] was solved similarly as follows.

$$k_\perp^2\phi - \frac{\partial^2\phi}{\partial z^2} = \frac{\omega_{p,dep}^2}{\omega^2}k_\perp^2\phi(\delta(z) + \delta(z-a)) \Rightarrow$$

$$k_\perp^2\phi + \lambda^2\phi = \frac{\omega_{p,dep}^2}{\omega^2}k_\perp^2\phi_o + \frac{\omega_{p,dep}^2}{\omega^2}k_\perp^2\phi_a exp(-i\lambda a) \Rightarrow$$

$$\phi(k_\perp^2 + \lambda^2) = \frac{\omega_{p,dep}^2}{\omega^2}k_\perp^2\phi_o + \frac{\omega_{p,dep}^2}{\omega^2}k_\perp^2\phi_a exp(-i\lambda a) \Rightarrow$$

$\phi = \frac{\omega_{p,dep}^2}{\omega^2(k_\perp^2+\lambda^2)}k_\perp^2\phi_o + \frac{\omega_{p,dep}^2}{\omega^2(k_\perp^2+\lambda^2)}k_\perp^2\phi_a exp(-i\lambda a) \Rightarrow$ The inverse transform is minimally complicated by the presence of the exponential term, the general machinery remains the same however.

$$\phi = \frac{1}{2\pi}\frac{\omega_{p,dep}^2}{\omega^2}\phi_o\int_{-\infty}^{\infty}\frac{k_\perp^2}{(k_\perp^2+\lambda^2)}exp(i\lambda z)d\lambda + \frac{1}{2\pi}\frac{\omega_{p,dep}^2}{\omega^2}\phi_a\int_{-\infty}^{\infty}\frac{k_\perp^2}{(k_\perp^2+\lambda^2)}exp(i\lambda(z-a))d\lambda$$

which leads to the solution

$$\phi(z) = \frac{\omega_{p,dep}^2}{2\omega^2}k_\perp\phi_0 exp(-k_\perp|z|) + \frac{\omega_{p,dep}^2}{2\omega^2}k_\perp\phi_a exp(-k_\perp|z-a|)$$

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